Application note 029: First Steps

This note is for someone who has just picked up a C47 or R47 and wants to get comfortable with it. It goes from the very first keystroke, through the settings most people like to make, and then on a short tour of what the calculator can really do: complex numbers, the equation solver, function graphing, statistics, and keystroke programming. It closes with two practical skills, saving your work and building a menu of your own. No prior calculator experience is assumed, and we teach the RPN way of entering numbers from scratch.

The C47 and R47 run the same firmware and work in almost exactly the same way. C47 is the firmware for the SwissMicros DM42 and DM42n; a C47 bezel, sold separately, relabels a DM42n's keys to match. The R47 is a separate SwissMicros calculator with its own keyboard, running the same firmware family. Because the firmware is shared, everything about how numbers, menus, and functions behave is identical; only the placement of a few keys differs. This note gives keystrokes for the C47, and a full map of each keyboard appears on pages 15 and 16.

A word on notation. A keystroke drawn as [5] means press that key. A first-shifted function is drawn with an orange marker, as in [f] [MODE]: on the C47 you press the yellow shift key once, then the key whose orange label is MODE. A second-shifted function is drawn in blue, as in [g] [CLK]: on the C47 you press the yellow shift key twice. The R47 has two physical shift keys instead, a yellow f and a blue g, so you press the matching one. The six keys under the display are soft keys; the screen prints a label above each, and a menu choice is drawn as a shaded key, [DEG]. To back out of anything, press [EXIT].

The pictures of the screen throughout are captures from the desktop simulator, taken at factory settings by running the very keystrokes printed beside them, so what you see should match what you get. One difference is worth knowing before it puzzles you: the tiny S above L in the top right corner is the simulator's stack-lift indicator, and on real hardware that corner carries the battery gauge instead.

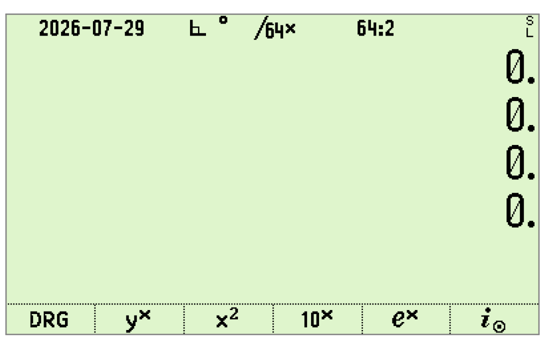
Contents

Revision	Date	Description
0.1	2026-07-22	First draft against firmware 0.109.03.04, circulated for comment.

1 First switch-on

Turn the calculator on with EXIT, which doubles as the ON key. The desktop simulator opens to the same screen. Loading the firmware onto real hardware is covered in Application Note 0 [1]; here we assume the calculator is already running C47 or R47.

The display has three parts. The top line is the status bar. On a new calculator it carries the date, a right-angle mark for rectangular complex mode, a degree sign for the angle unit, the largest fraction denominator, the integer word size and base, and a small f or g when a shift is pending. Nearly every one of those is switchable, so yours may show more or fewer of them. The middle shows the stack, where your numbers live. The bottom line, when a menu is open, shows the six soft-key labels; with no menu open it shows MyMenu, which we come back to at the end.



A new calculator, straight out of a reset: an empty stack and MyMenu along the bottom.

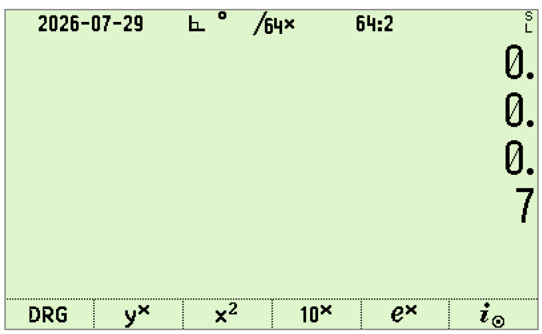
Two keys rescue you at any time. [EXIT] backs out of a menu or an entry, and the backspace key, [←], rubs out the last digit; when you are not typing it clears the number in the display. If you feel lost, press [EXIT] a few times to get back to the plain stack.

2 The RPN way

The C47 and R47 use RPN, Reverse Polish Notation. There is no equals key. You put numbers on a small stack of registers and then press an operation, which acts on the numbers already there. It feels unusual for about ten minutes and then feels faster than anything else.

The stack holds eight numbers out of the box, and the bottom four, named X, Y, Z, and T from the bottom up, do nearly all the work. X is the number in the display. The [ENTER] key pushes a copy of X up into Y so you can type a second number without the two running together. To add three and four: type [3], press [ENTER] to lock it in, type [4], then press [+].

[3] [ENTER] [4] [+] → `X: 7`



Three plus four, the RPN way. The answer lands in X, the bottom line.

The plus takes the two numbers and leaves their sum in X. You never said where the answer should go; it just appeared. The stack shines when a calculation has several steps, because intermediate results wait on it and are used automatically. To add 2, 3, and 4:

[2] [ENTER] [3] [ENTER] [4] [+] [+] → `X: 9`

The first plus makes 7 from 3 and 4; the second adds that to the 2 waiting underneath. A few more keys finish everyday arithmetic. The change-sign key, [+/−], flips the sign of X. The square-root key works in place:

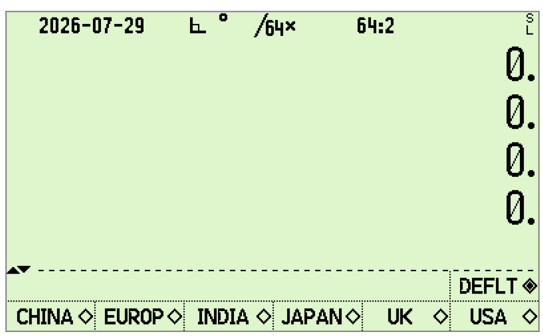
[2] [$\sqrt{x}$] → `X: 1.414213562...`

The x⇄y key swaps X and Y when you enter two numbers in the wrong order, and f LASTx brings back the value that was in X before the last operation, which rescues you after a slip. That is the whole idea: type a number, press ENTER only when you must separate it from the next, and press operations as you go.

3 Settings worth making first

Almost everything is adjustable, and the defaults are sensible, so change as little or as much as you like. The settings below are the ones most people reach for on day one. The full tour of configurability is Application Note 28 [13], and saving a bundle of settings to recall later is Application Note 15 [9]. You need not save anything by hand for ordinary use: the calculator writes its state to a backup file at power-off and restores it at power-on.

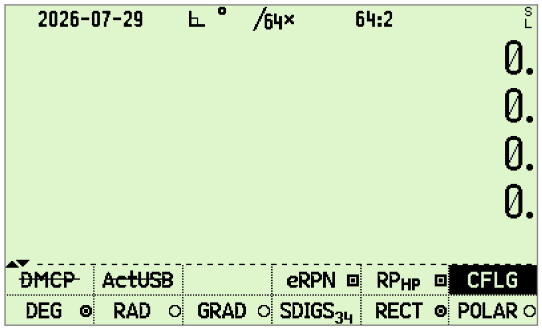
The single most useful first step is to set your region. Open the display menu with f DISP and page to the third page, where you will find CHINA, EUROP, INDIA, JAPAN, UK, and USA; the shifted row above them holds DEFLT, the factory choice. The menu has three pages and the paging wraps, so a single press of the down-arrow, ↓, is the quickest way there. One press of a region key retunes the regional conventions together, the decimal mark and the digit grouping among them; Application Note 28 [13] lists everything it touches and why the conventions differ.



The third page of f DISP: one press retunes the regional conventions together.

Next, choose how many digits you see. Still under f DISP, the first row offers FIX, SCI, ENG, UNIT, SIG, and ALL. FIX shows a fixed number of decimals: press FIX then a digit, so FIX 4 shows four places and displays π as `3.141 6`, the space being the digit grouping you chose above. SCI is scientific notation, ENG is engineering notation in powers of a thousand, and ALL shows every significant figure.

Pick your angle unit before any trigonometry. Open the mode menu with f MODE and choose DEG, RAD, or GRAD; the small circle beside each label is filled in on the one in force. It matters at once: in degrees, the sine of 30 is exactly one half.



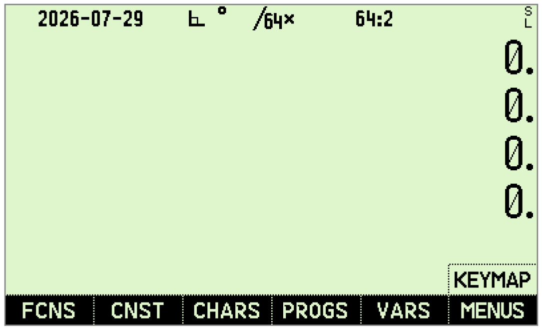
The first page of f MODE, with DEG in force. A label with a line struck through it, like DMCP and ActUSB here, is a function the build you are running does not offer; these two need real hardware, so the simulator crosses them out.

Degrees-minutes-seconds and multiples of π live on the trig menu, g TRG. The date and time formats are on the second page of the clock menu, g CLK: page to it and pick DMY, MDY, or YMD, and CLK24 or CLK12. Two more that some people like: the second page of f MODE switches between the eight-level stack you start with, SSIZE8, and the classic four-level stack, SSIZE4, and if you work in electronics and prefer j for the imaginary unit, the second page of f DISP offers CPXi and CPXj.

4 Finding your way around the menus

There are more functions than keys, so most functions live in menus. A menu fills the bottom of the screen with up to six labels, one above each soft key; press the soft key to run it. Many menus run to several pages, and the up and down arrows page through them. Each label can carry a shifted function too, shown on the lines above it and reached with a shift before the soft key.

When you cannot remember where something is, the catalog finds it. f CAT opens alphabetical catalogs of every function, menu, variable, and program: FCNS for functions, CNST for physical constants, CHARS for characters, PROGS for your own programs, VARS for your variables, and MENUS for the menus themselves. Pick a group and read down the list. Selecting an item runs it, or, while you are typing a program, inserts it.



f CAT and its six groups. Everything the calculator can do is in here somewhere.

The complete keyboard, key by key, is mapped on pages 15 and 16, and Application Note 18 [10] covers it in detail.

5 A short tour

The next few pages show things a plain calculator cannot do. Try them in order.

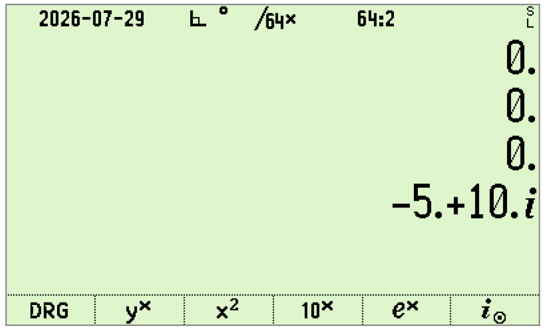
5.1 Complex numbers

The calculator treats complex numbers as ordinary values. A switch called CPXRES decides whether a real function that would leave the real line returns a complex number instead of an error. It is already on when the calculator is new; if you ever find it turned off, put it back with f MODE, second page, CPXRES. The quickest way to see which way the switch is set is simply to use it. Take the square root of minus one:

1 +/− $\sqrt{x}$ → `X: 1.i`

That is i, the imaginary unit, which no real-only calculator will give you; a pure imaginary is shown as `1.i`, with no real part written. To type a complex number directly, enter the real part, press the i operator (the blue function on the sine key), and enter the imaginary part. Multiplying $(1 + 2i)$ by $(3 + 4i)$:

1 i 2 ENTER 3 i 4 × → `X: -5.+10.i`



$(1 + 2i)(3 + 4i)$ worked out on the stack. Complex values sit in the registers exactly as real ones do.

which is right, since $(1 + 2i)(3 + 4i) = 3 + 10i + 8i^2 = -5 + 10i$. There is also a f COMPLEX key that joins two stack numbers into one complex value, and a whole g CPX menu of complex tools. Complex numbers, rectangular and polar, are the subject of Application Notes 1 and 10 [2, 6].

5.2 The equation solver

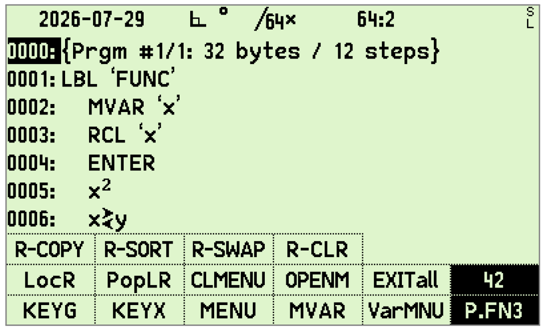
The solver finds the value of a variable that makes an equation true, that is, it finds roots. You give the calculator your function as a short program, hand the solver two guesses, and it hunts down the answer. Let us solve $x^2 + 2x - 2 = 0$.

First write the function as a program named FUNC. Declaring the variable with MVAR is what lets the solver later ask you for it, and lets the grapher plot it.

```
0001: LBL 'FUNC'
0002:   MVAR 'x'
0003:   RCL 'x'
0004:   ENTER
0005:   x²
0006:   x⇄y
0007:   2
0008:   ×
0009:   +
0010:   2
0011:   -
0012: END
```

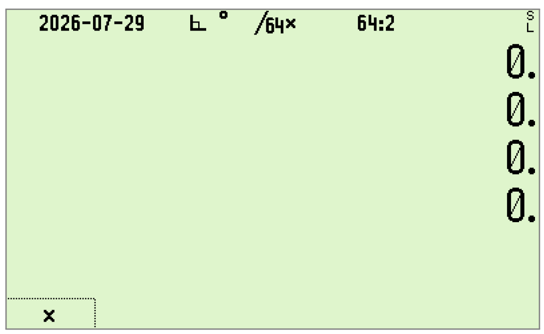
To key it in, open a new program with XEQ . . and switch to program-entry mode with f PRGM. Start the label with g LBL, then f α, type F U N C, and ENTER. Declare the variable next. MVAR lives two pages into the program-functions menu that is now on the screen: press P.FN1, then P.FN2, then MVAR, and type f α × ENTER. Now key the function: RCL f α × ENTER, then ENTER, f x^2, x⇄y, 2, ×, +, 2, −. Leave program-entry mode with f PRGM.

While you are still in program-entry mode the screen lists what you have typed, so you can check it against the listing above before you leave. The up and down arrows walk the listing.



FUNC keyed in, seen from the top. The header line counts the bytes and the steps.

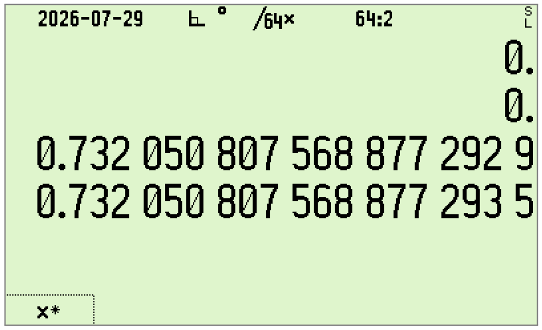
Now solve it. Open [f] [ADV], choose [SOLVE], and name the program with [f] [α] [F] [U] [N] [C] [ENTER]. The variable x now appears as a menu key, and it is the only one, because FUNC declared only that one [MVAR].



After naming the program, its variable appears as a soft key. This one menu key does all three jobs: it takes the first guess, the second, and then the solve.

Give a guess below the root and one above it, pressing the x key after each, then press it once more, with nothing typed ahead of it, to solve:

[2] [+/−] [x] [2] [x] [x] → `x: 0.732050807568877...`



The root, found. The star on the menu key marks the variable that was solved for.

which is $-1+\sqrt{3}$, one of the two roots. Guide it toward the other with more negative guesses, say -11 and -10, and it returns $-2.732050807568877\ldots$, which is $-1-\sqrt{3}$. Each solve returns one root; you steer it to the next by changing the guesses, exactly as on the classic HP machines. Full worked examples, including the algebraic f EQN route where you type the formula as on paper, are in Application Notes 11 and 22 [7, 11].

5.3 Graphing a function

The calculator draws graphs on its own screen, and it is worth seeing it do something beautiful. Let us plot a wave packet, $f(x) = \sin(5x)\, e^{-x^2/8}$: a burst of oscillation held inside a smooth bell-shaped envelope, the shape of a pulse of light or a plucked string fading to silence. It is a handful of keystrokes and it shows off the screen.

Switch to radians first with f MODE, RAD. Then enter the function as a program, the same way you built FUNC:

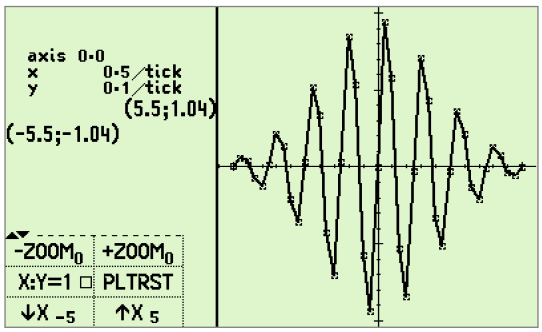
```
0001: LBL 'WAVE'
0002:   MVAR 'x'
0003:   RCL 'x'
0004:   5
0005:   ×
0006:   SIN
0007:   RCL 'x'
0008:   x²
0009:   8
0010:   ÷
0011:   CHS
0012:   e^x
0013:   ×
0014: END
```

To key it in, open a new program with XEQ . ., f PRGM, then g LBL f α W A V E ENTER, and declare the variable as before: P.FN1, P.FN2, MVAR, then f α × ENTER. Then key the function: RCL f α × ENTER, 5,

×, SIN, then RCL f α × ENTER, f x^2, 8, ÷, +/−, f e^x, ×. Leave with f PRGM. Now plot it: f ADV, PLTf, name it f α W A V E ENTER, and the variable x appears as a menu key, just as it did for the solver. Give it the two ends of the range, pressing the x key after each, then press it once more to draw:

5 +/− x 5 x x → `plots WAVE,` $-5 \le x \le 5$

The screen fills with a train of waves swelling to a peak in the middle and dying away smoothly to each side, with a zoom menu below. Press EXIT when you are done looking.



$\sin(5x)\, e^{-x^2/8}$ over $-5 \le x \le 5$. The corners give the range covered, the tick spacing, and where the axes cross. The menu underneath zooms in and out, forces a square aspect, and resets the plot.

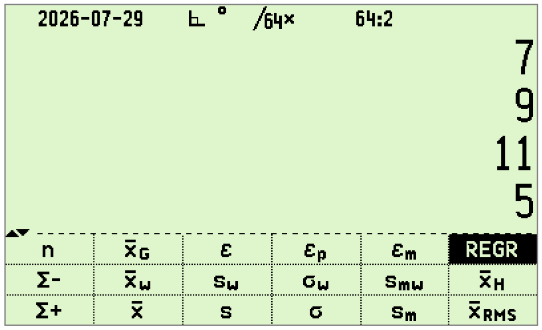
Change the 5 to a larger number for faster wiggles, or the 8 for a wider envelope, and plot again. Plotting, including scatter plots of data and curves in the complex plane, is covered in Application Notes 3, 13, and 22 [3, 8, 11].

5.4 Fitting a line to data

The calculator is a quietly powerful statistics machine. Suppose you measured a quantity y at five settings of x and recorded the pairs $(1, 3)$, $(2, 5)$, $(3, 7)$, $(4, 9)$, $(5, 11)$. Enter each pair with the Σ+ key at the top left: key the y value, ENTER, the x value, then Σ+.

3 ENTER 1 Σ+ 5 ENTER 2 Σ+ … 11 ENTER 5 Σ+ → `5 points`

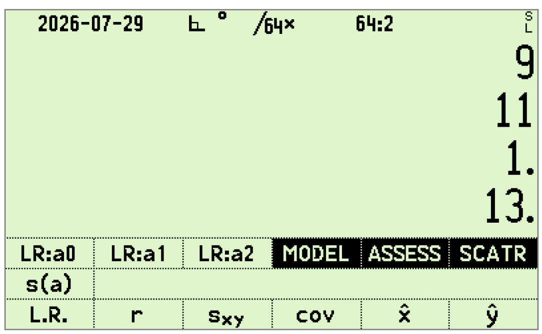
Now open the statistics menu, f STAT, and read your data off it. $\bar{x}$ gives the mean of the x values, 3, with the mean of the y values just above in Y, and s gives the sample standard deviation, 1.5811; the σ key beside it gives the population version, 1.4142.



f STAT with the five pairs entered. The last four y values are still on the stack behind the menu.

Open the regression submenu: press the shift key twice, then the REGR soft key. There r reports the correlation, 1: the points lie on a perfect straight line. Its intercept and slope sit on the blue row of the same menu, so press shift twice before each: LR:a0 is 1 and LR:a1 is 2, so the line is $y = 2x + 1$. Best of all, you can predict. Key a new x and press $\hat{y}$; at $x = 6$ the fitted line gives

6 $\hat{y}$ → `y: 13`



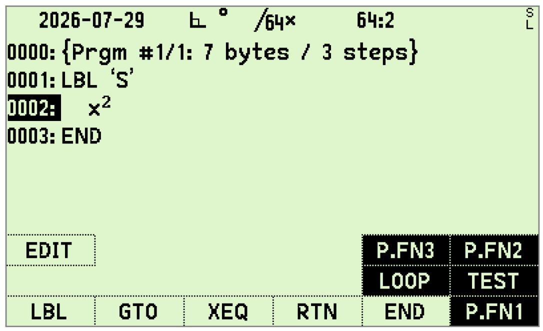
The regression menu, and the line's prediction at $x = 6$.

Real data is rarely this tidy, and that is where the rest of the menu earns its keep: the correlation tells you how well a line fits, and a MODEL choice fits curves, logarithms, exponentials, and powers just as easily. Curve fitting and statistical plots are covered in Application Notes 3 and 13 [3, 8].

5.5 Writing a small program

A program is a recorded sequence of keystrokes with a name, and writing one is just doing the calculation by hand in program-entry mode. Let us make a program S that squares X. Open a fresh program with XEQ . . as before, press f PRGM to enter program-entry mode, key the label and the one step that does the work, then press f PRGM again to leave. The g LBL key starts a label, the f α key switches to letter entry so you can type the name, and ENTER finishes the name.

XEQ . . f PRGM g LBL f α S ENTER f x^2 f PRGM → 0001: LBL 'S'



The whole program: a name, one step of work, and the END the calculator adds for you.

Now put a number in X and run the program by name:

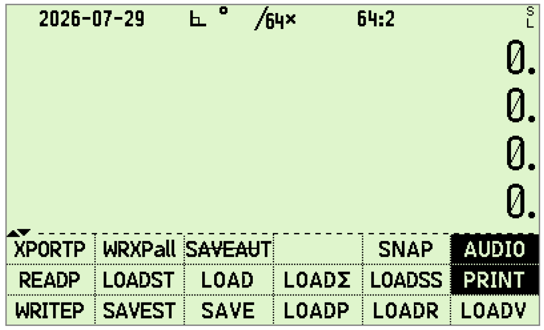
5 XEQ f α S ENTER → X: 25

The program squared 5 to give 25. Anything you can do by hand you can record this way and replay it, or assign it to a key so it runs with one press. Programs can also loop and decide, which turns the calculator into a small computer: the Σn function on the f ADV menu runs a labeled program once per term and adds the results, so summing the squares from 1 to 3 gives $1 + 4 + 9 = 14$. Programmed sums are the subject of Application Notes 8 and 9 [4, 5].

6 Saving and restoring your work

Your work is safe without any effort on your part. When you switch the calculator off it writes its whole state, the stack, your variables, your programs, and every setting, to a backup file, and reads it all back when you switch on. You never lose your place, even across a battery change.

You can also take a named snapshot to keep, which is worth doing before a big experiment. Open the input and output menu with g I/O and press SAVEST; the calculator opens its file screen, where you give the snapshot a name such as MYWORK and confirm with ENTER. The whole state is written to a file of that name. Now tinker as freely as you like; when you want it all back, open g I/O again, press the shift key once and then LOADST, which sits on the orange row just above SAVEST, and choose your file. Everything returns exactly as it was, down to the number in X.



The g I/O menu. SAVEST writes a state file and LOADST, directly above it on the orange row, reads one back.

7 Making the calculator your own

You can gather the functions and programs you use most into a menu of your own, called MyMenu, so they are always a press away instead of buried in a catalog. You have seen MyMenu already: it is the menu sitting at the bottom of the screen whenever nothing else is open, stocked from the factory with six starters, DRG, y^x, x^2, 10^x, e^x, and the polar i. Press EXIT until the other menus close and it reappears. Those starters are yours to overwrite.

Say you want your squaring program S and the sine function on it. Press f ASN, which is the yellow shift then the 1 key, to begin an assignment. Choose what to assign: for the program, open the catalog with f CAT, choose PROGS, and pick S; for a plain function like SIN, just press its key. Then press f USER, which folds the menus away and brings MyMenu back, and press the soft key where you want the item to sit. Repeat for each one. From then on, those soft keys run your chosen items.

A long press of the yellow shift key opens the HOME menu of common functions; on the third page of f MODE you can point that long press at MyMenu instead, and choose which of the two greets you as the resting menu. The same f ASN command assigns anything, a function, a program, a variable, or a whole menu, to any key you like, so you can shape the entire keyboard around your work. Assignments, the four editable menus, and the ready-made ribbons of common functions are covered in Application Note 18 [10].

8 Where to go next

You now have the essentials: RPN entry, the settings that make the calculator yours, the menu system, and a first look at complex numbers, solving, graphing, and programming. Application Note 18 [10] is the full keyboard reference, Application Note 28 [13] tours everything you can configure, and if you come from an HP-42S, Application Note 26 [12] explains what is compatible and what is new. The complete function reference lives at 47calc.com [14], alongside the project wikis [15, 16]. Most of all, experiment: the backup keeps your state safe, EXIT always backs you out, and nothing you press survives a fresh start.

The C47 keyboard

Primary function on the key; orange = f (shift once) above; blue = g (shift twice) below.

→I	y^x	x^2	10^x	e^x	α
Σ+	$1/x$	$\sqrt{x}$	LOG	LN	XEQ
a b/c	#	.ms	.d	LBL	GTO
$\lvert x\rvert$	%	π	ASIN	ACOS	ATAN
STO	RCL	R↓	SIN	COS	TAN
∡	Δ%	$\sqrt[x]{y}$	i	→R	→P

COMPLEX	LASTx	MODE	DISP	UNDO
ENTER	x⇄y	+/−	EEX	←
CPX	STK	TRG	EXP	CLR

BST	EQN	ADV	MATX	STAT
↑	7	8	9	÷
REGS	HOME	FIN	X.FN	PLOT
SST	BASE	CONV	FLAG	PROB
↓	4	5	6	×
FLGS	BITS	CLK	REAL	INTS
	ASN	USER	P.FN	PRINT
f / g	1	2	3	−
	KEYS	α.FN	LOOP	I/O
OFF	VIEW	SHOW	PRGM	CAT
EXIT	0	·	R/S	+
SNAP	STOPW	INFO	TEST	CNST

The R47 keyboard

The R47 has two shift keys: a yellow f and a blue g. Primary on the key; f above, g below.

References

[1] C47/R47 Application Note 000, *Installation*.
[2] C47/R47 Application Note 001, *Complex number engineering example*.
[3] C47/R47 Application Note 003, *Graphing*.
[4] C47/R47 Application Note 008, *Programmed Sums 1*.
[5] C47/R47 Application Note 009, *Programmed Sums 2*.
[6] C47/R47 Application Note 010, *Complex solver continued*.
[7] C47/R47 Application Note 011, *EQN and ADV menu, solver and integrator*.
[8] C47/R47 Application Note 013, *Graphs*.
[9] C47/R47 Application Note 015, *Write and Read Setting Groups*.
[10] C47/R47 Application Note 018, *Keyboard*.
[11] C47/R47 Application Note 022, *RPN solve, integration and plot*.
[12] C47/R47 Application Note 026, *42S Compatibility*.
[13] C47/R47 Application Note 028, *Configurability Tour*.
[14] C47/R47 function reference. `https://47calc.com/documentation`
[15] C47 developer wiki. `https://gitlab.com/rpncalculators/c43/-/wikis/home`
[16] C47 community wiki. `https://gitlab.com/h2x/c47-wiki`

Author

David Emerson, d@nnix.com